

Here's a summary of what was built:

Backend — FastAPI + SQLite (`main.py`)

- On startup it reads your CSV, drops and recreates the `interactions` table, and inserts rows in batches of 10,000 — so even 200k+ rows loads in a few seconds
- Two indexes (one per drug-ID column) make per-drug lookups fast regardless of DB size
- Three endpoints: `/health` (connection check + row count), `/search` (autocomplete), `/regime/risk` (the main scorer)
- Drug lookup accepts IDs **or** names, case-insensitively

Risk model:

- Each drug's risk = its average interaction strength across all rows where it appears (as either drug A or B)
- Regime risk = sum of those averages
- Coverage = % of the regime's unique drug pairs that have at least one recorded interaction

Frontend — React (`App.jsx`)

- Tag-style drug input with live autocomplete (debounced, 200 ms)
- Keyboard navigation (arrows, enter, backspace to remove last tag)
- Results show: total risk, DB coverage %, per-drug risk bars colour-coded green → amber → red, and a warning for any drugs not found in the DB

To run:

```
bash
cd backend
pip install -r requirements.txt
cp /your/data.csv interactions.csv
uvicorn main:app --reload --port 8000
```

Then use the JSX in any Vite/CRA project, or paste it into the Claude artifact runner for a live preview. The README covers all the details.

Here's a step-by-step guide for getting this running on Windows in VSCode.

Prerequisites — install these first

1. **Python 3.11+** — download from python.org. During install, tick "**Add Python to PATH**"
 2. **Node.js 18+** — download from nodejs.org (needed for the React frontend)
 3. **VSCode** — with the **Python** and **ES7+ React** extensions recommended
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1. Set up the project folder

Open a VSCode terminal (`<ctrl>+<`>`) and create your structure:

```
mkdir drug-risk
cd drug-risk
mkdir backend frontend
```

Save `main.py` and `requirements.txt` into `backend/`, and `App.jsx` into `frontend/`.

2. Create a Python virtual environment for the backend

```
bash
cd backend
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
```

You should see `(venv)` in your terminal prompt. VSCode may ask if you want to use this environment — click **Yes**.

3. Add your CSV

Copy your interactions CSV into the `backend/` folder and name it `interactions.csv` — or set the path via an environment variable:

```
bash
set DRUG_CSV=C:\path\to\your\file.csv
```

4. Run the backend

From inside `backend/` with the `venv` active:

```
bash
uvicorn main:app --reload --port 8000
```

You should see something like:

```
[info] Loaded 200,000 interactions into 'interactions.db'.
INFO: Uvicorn running on http://127.0.0.1:8000
```

Leave this terminal running. Open a **new terminal** for the frontend.

5. Set up the React frontend (Vite)

```
bash
cd ../frontend
npm create vite@latest . -- --template react
npm install
```

When prompted, choose **React** and **JavaScript**.

Replace the generated `src/App.jsx` with the `App.jsx` file from this project. Then run:

```
bash
npm run dev
```

Open <http://localhost:5173> in your browser.